# Smart Guardian
# A Project for Community Safety

**Presented by : Team TechOps**

Team Members: TamilSelvi SV , Surya Narayanan S, Kaaviya BS, Ashwin Kumar VM

Educational Institution: Vellore Institute of Technology Chennai, Tamil Nadu

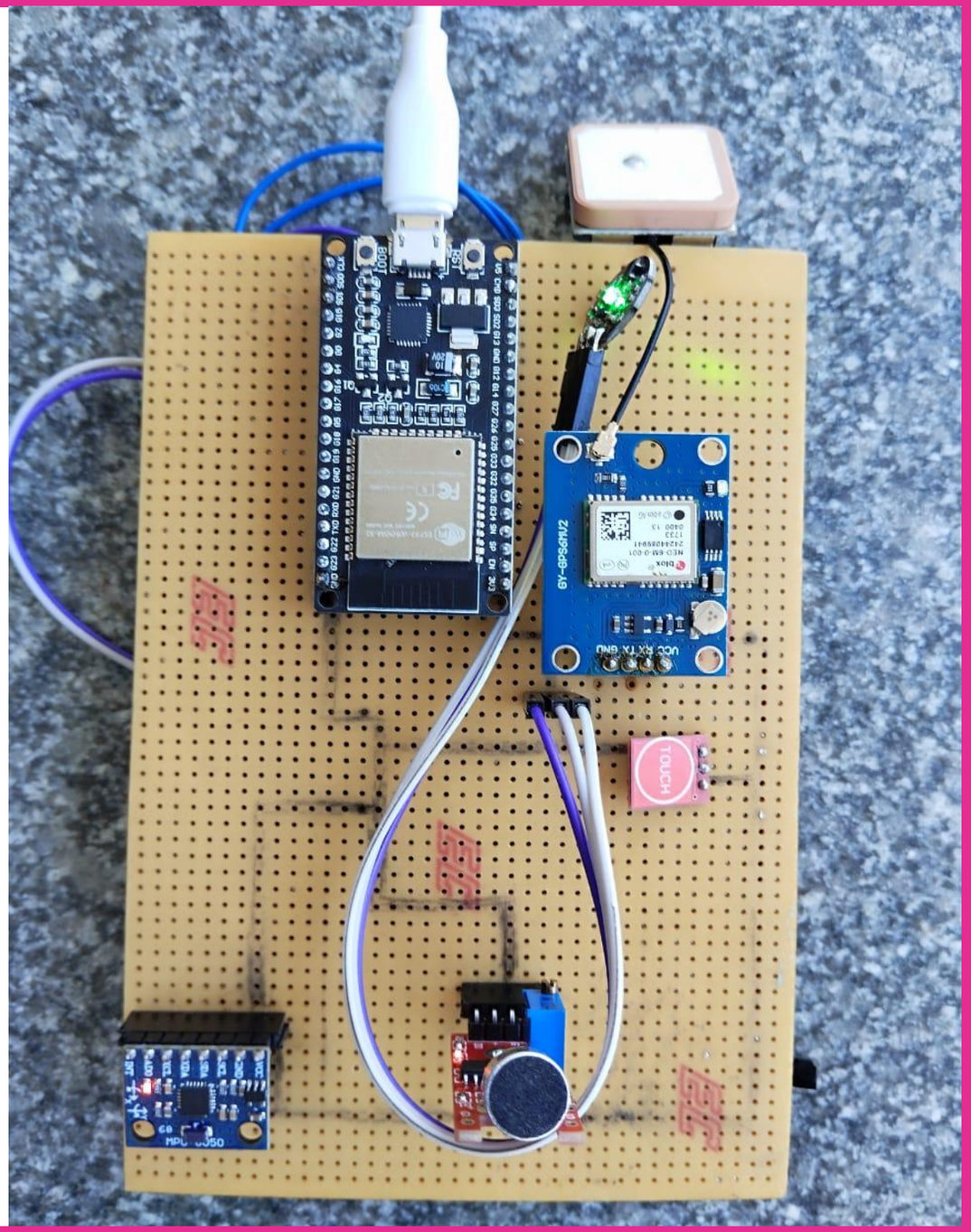
Project Category: Community Empowerment & Public Safety

# Abstract

Smart Guardian is a wearable technology project designed to address critical needs in elderly fall detection and women's safety. The project leverages a combination of advanced sensors including the MPU6050, Touch Sensor, Sound Sensor, ESP32,ESP32 cam, Heart Rate Sensor, and GPS. By providing timely assistance and enhancing personal security, Smart Guardian aims to empower individuals and foster a safer community.

- Focus: Addressing critical needs in elderly fall detection and women's safety using wearable technology.

- Impact: Providing timely assistance and enhancing personal security.



# The Problem: A Pressing Need for Safety

## Elderly Falls

Statistics: "Every year, 3 million older adults are treated in emergency departments for fall injuries" (CDC).

Risk Factors: Age-related physical changes, chronic conditions, medication side effects, and environmental hazards contribute to the high incidence of falls among the elderly.

Why : "Falls cost the U.S. healthcare system $50 billion annually" (CDC). This includes medical expenses, rehabilitation, and long-term care costs.

## Women's Safety

Statistics: "Nearly 1 in 3 women worldwide have experienced physical or sexual violence" (UN Women).

Risk Factors: Underreporting of incidents, lack of access to support services, and societal attitudes contribute to the ongoing threat to women's safety.

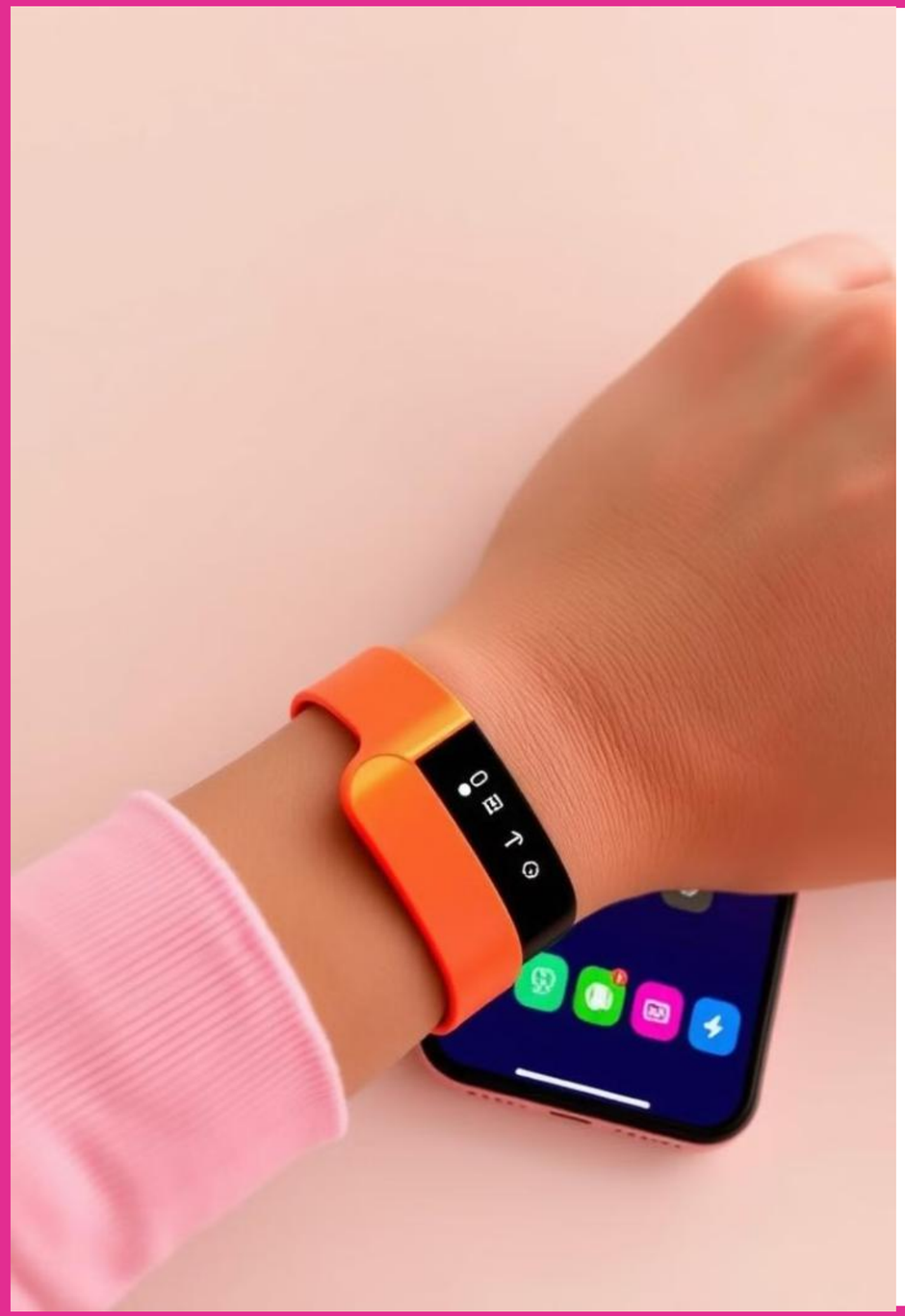
Why : "Personal safety devices can deter up to 70% of potential attackers”.

## Child Safety

Statistics: “Every year, over 800,000 children are reported missing

Risk factors: Lost child incidents peak during summer months and large public events.

Why: GPS tracking can reduce response time by up to 60%.



# Smart Guardian: The Solution

A multi-sensor wearable bracelet designed for proactive safety measures.

## Key Components

MPU6050 (Accelerometer/Gyroscope): Fall Detection Touch Sensor: User Input & Alert Trigger Sound Sensor: Ambient Sound Monitoring for Distress ESP32: Real-time Data Processing & Connectivity Heart Rate Sensor: Vital Sign Monitoring GPS: Location Tracking for Emergency Response

# Features: Tailored for Different Needs

## Elderly

Fall Detection: Automatic detection and alert system.
Consciousness Verification: Immediate follow-up to confirm user status.

## Women

Fake Call Activation: Discreetly trigger a simulated incoming call.
Location Tracking: Real-time GPS tracking for trusted contacts.
Sound Monitoring: Recording and analysis of surrounding sounds.

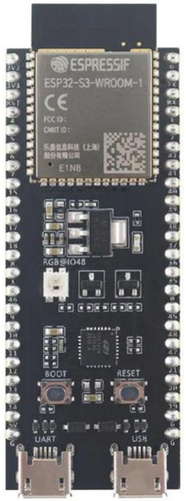
## Child Safety

GPS radius lock: Set a bounding radius around a central point (say your house) and send a notification if your child leaves the circle

# Tech Stack: Hardware & Software Integration

## Hardware

ESP32 microcontroller for data processing and wireless communication. Sensors: MPU6050, Touch Sensor, Sound Sensor, Heart Rate Sensor, GPS module. Power Source: Rechargeable lithium-ion battery with 72-hour life (based on internal testing)



## Software

Arduino IDE for firmware development. Custom mobile app (iOS and Android) for device management and alerts. Cloud-based data analytics platform “Firebase” for data storage and reporting.

Firebase

ARDUINO

# Impact Metrics: Measuring Success

## Elderly

Reduction in fall-related emergency response time: Target 50% decrease. Increased user confidence in independent living: Survey-based evaluation.

## Women

Reduction in reported incidents in monitored areas: Data collection from local authorities/police reports. Improved sense of personal security: Survey-based assessment.

## Overall

User adoption rate: Number of active users within the community. System reliability: Uptime and successful alert rate.

# Conclusion: Smart Guardian - A Safer Future, Today

Smart Guardian offers a comprehensive solution to enhance safety and empower individuals in the community. By integrating advanced technology, the project aims to reduce fall-related risks for the elderly and provide women with greater security. The project's success will be measured by a reduction in emergency response times, an increased sense of safety, and a higher adoption rate among community members. With continued development and community support, Smart Guardian has the potential to be a vital tool in creating a safer and more secure future for all.